

QUMIA

Quantitative Ultrasound Muscle Imaging Analysis software

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USING QUMIA

Image Collection and Storage

During each examination, **three images per muscle** are typically acquired. Ensure that all images are annotated with the **muscle name** and **measurement side** before export. After exporting, store the images in an accessible location, with **each examination placed in its own folder**. Only imaging files should be present in this folder; ensure no additional files are included.

Starting QUMIA

QUMIA can be used either through the **MATLAB interface** or as a **standalone application**.

- **Using the MATLAB Interface**
 - Start MATLAB.
 - Navigate to the QUMIA installation folder.
 - Launch QUMIA by either:
 - Right-clicking *QUMIA.mlapp* and selecting **Run**, or
 - Typing QUMIA in the Command Window and pressing Enter.
- **Using the Standalone Application**
 - Run **QUMIA.exe** from the QUMIA installation folder.

Beginning an Analysis

When QUMIA starts, all functionality is initially disabled. Features become available once data has been loaded.

- To begin a new analysis: **Menu → New analysis**
- To continue or review a previous analysis: **Menu → Continue analysis**

An analysis consists of all images belonging to a **single examination**.

Loading Examination Data

- When starting a new analysis, a dialog window appears.
- Navigate to the directory containing all examinations.
- Select the folder containing the images for one examination and click **Open**.

Entering Participant Characteristics

Enter all required participant details in the dialog window. Some fields may already be populated if information was entered during scanning. Click **Save & continue** to proceed.

Participant characteristics can be edited later using the **Patient characteristics** button.

Annotating and Analysing Images

All loaded images appear in the **Image** list on the right-hand side. Regardless of filename, QUMIA assigns them sequential numbers. Selecting an item displays the corresponding image.

The following steps can be completed in any order:

- **Annotating Muscle and Measurement Side**
 - Use the **Muscle** and **Side** boxes to annotate each image.
 - This information is not stored in the DICOM header and must be entered manually.

- The image name in the list updates automatically.
- The muscle list is derived from *Muscles.xlsx*.
- The option **No analysis** is available for images that should be excluded from the report.
- **Drawing a Region of Interest (ROI)**
 - Click **Add** in the ROI box to draw a polygonal ROI using MATLAB's polygon tool.
 - Each click adds a vertex; close the ROI by clicking the first point or double-clicking.
 - Redraw the ROI by clicking **Add** again, or remove it with **Remove**.
 - The ROI summary statistic (see *Setting up QUMIA*) is displayed automatically and cannot be edited.
- **Measuring Muscle Thickness**
 - Click **Add** in the Thickness box to draw a measurement line.
 - Click once to place the first point, drag to the second point, and release.
 - Remove the line using **Remove**.
 - The calculated muscle thickness (cm) is displayed and **can be edited**.
 - If thickness was measured during scanning, you may enter the value manually and remove the line.

Continue until all images have been fully annotated.

Report Preview

Select **Report preview** to open a new window displaying:

- Observed values (ROI summary statistic and muscle thickness), averaged across images of the same muscle and side
- Predicted values and z-scores when normative models are available
- Optional display of individual measurements (toggle **Show individual measurements**)

Saving an Analysis

When the analysis is complete, select **Menu → Save**. A dialog window appears during the save process—wait until it closes automatically.

QUMIA restructures the examination folder into three new subfolders:

- **DICOM images**
 - All original DICOM files are copied here with new filenames based on muscle and side annotations.
- **Images**
 - Contains .tif images with ROI and thickness overlays.
 - Filenames match those in the DICOM folder.
- **Saves**
 - **Analysis results.xlsx** — summary report, detailed data, and participant information
 - **Analysis save.mat** — MATLAB data file
 - **bodyChart.jpg** — schematic body with echogenicity z-scores

Continuing an Existing Analysis

To revisit or continue an analysis:

- Start QUMIA.
- Select **Menu → Continue analysis**.

- Choose the **parent folder** containing the three subfolders (*DICOM images, Images, Saves*).
- Click **Open**.

Normative Data Integration

QUMIA supports integration of normative models (see *Creating normative values*) and automatic calculation of z-scores for echogenicity and thickness.

The **Normative models** button opens a window where:

- Muscles with available models are listed
- Models can be visualised
- The **Individual BMI – Average BMI** toggle switches between personalised and smoothed model visualisation

INSTALLATION

Setting up your Ultrasound device

Before beginning data collection, it is essential to correctly configure your ultrasound device. Without proper standardisation, echogenicity measurements will not be valid, and QUMIA analysis results cannot be used reliably. We therefore strongly recommend contacting us for remote support during this process.

A general principle in quantitative muscle ultrasound is that the image must be **reproducible**, even if it is not aesthetically pleasing. The following guidance will help you establish appropriate settings. Begin with an MSK preset and adjust the parameters as required.

- **Probe and preset:** Use an MSK preset with a linear probe operating between 10–15 MHz.
- **Depth:**
 - For most muscles, a depth of **4 cm** is sufficient.
 - For adult upper-leg muscles, **6 cm** is often required.
 - For facial muscles, **2 cm** is typically adequate.
 - For children, **4 cm** generally works for all muscles.
 - Because scanning depth has a substantial effect on echogenicity, determine the required depth for each protocol in advance. Create multiple presets if necessary.
- **Post-processing functions:** Disable all manufacturer-specific post-processing features wherever possible. These settings often provide only categorical options (e.g., “low”, “medium”, “high”, “off”), while basic ultrasound parameters cannot be fully disabled.
- **Gain map (‘map’):** Select a gain map that is as linear and as true-grey as possible. Although this is difficult to judge visually, it can be objectively assessed using greyscale-scale (wedge) analysis.
- **Image homogeneity:** Aim to homogenise the image to minimise depth-dependent variation in echogenicity.
 - Start with all TGC sliders in the neutral position (and lateral gain sliders, if available).
 - If physical sliders are not present, equivalent controls will be available in the software interface.
 - Scan a homogeneous medium such as an ultrasound phantom or a block of tofu.
 - Adjust focus settings (focus depth, number of foci, spacing between foci). Complete removal of depth dependency is not possible. If you choose to include a specific TGC configuration in your protocol, ensure it can be reproduced consistently.
- **Dynamic gain (‘compression’):** Use a dynamic gain setting between **60 and 90 dB**.
- **Overall gain:** Select a gain level at which:
 - Muscle tissue in a healthy participant is **not saturated to black** (target mean echogenicity: 60–80).
 - Muscle tissue in a severely affected patient is **not saturated to white**.

Exporting Images

Establish a workflow for exporting images. QUMIA requires images in **DICOM format** stored in an accessible location. Depending on the centre:

- Images may be exported automatically to a designated folder in parallel with storage on the imaging server, or
- Manual export (e.g., via USB flash drive) may be necessary.

Setting up QUMIA

QUMIA is developed in MATLAB but can be used on systems **with or without** a MATLAB installation.

- When MATLAB is available, QUMIA can be run directly through the MATLAB interface.
- When MATLAB is not installed, the **standalone application** must be used. Some centres choose the standalone version regardless of MATLAB availability.

During installation, several steps require running MATLAB code. Equivalent results can be achieved using alternative methods if preferred. Remote support is available on request.

Installation Using the MATLAB Interface

- Create a designated installation folder in your preferred location.
- Copy all contents of the \Source\ folder into the installation folder.
- Copy all contents of the \Sitefiles_blank\ folder into the installation folder.

Installation Using the Standalone Application

- Create a designated installation folder in your preferred location.
- Copy all contents of the \Standalone\ folder into the installation folder.
- If MATLAB is **not** installed on the machine, install the free MATLAB Runtime. See [Download and Install MATLAB Runtime - MATLAB & Simulink](#) for instructions.
- Copy all contents of the \Sitefiles_blank\ folder into the installation folder.

Editing QUMIA_config.txt

Do **not** change the filename. QUMIA relies on this file to operate correctly. Avoid using spaces before or after the = sign for each attribute.

- **defaultFolder:** Defines the default directory shown when starting a new analysis. Use a folder *other than* the installation directory for exported imaging data. Example: C:\UltrasoundData (no trailing backslash).
- **outputFolder:** QUMIA generates an .xml file for potential import into electronic patient records. Specify a full path, ideally different from both defaultFolder and the installation folder.
- **muscleList:** Full path (including filename and extension) to an Excel file listing all muscles measured in your centre. Example: C:\Program Files\QUMIA\Muscles.xlsx.
 - Cell A1 must contain the word *Muscle*.
 - Each subsequent cell in column A should contain a muscle name.
- **Margin:** Defines the lateral margin excluded from echogenicity analysis. Due to beam compounding, the outer edges of the image contain less data. Enter a value between **0 and 0.5**. Default: **0.15** (15% removed from each side).
- **Depth:** Used internally to match ultrasound presets based on scan depth. QUMIA calculates depth from DICOM header information. For each preset, calculate the depth once using MATLAB:
 - `info = dicominfo('fullPathToDICOMFile');`
 - `iSOUR = info.SequenceOfUltrasoundRegions.Item_1;`
 - `depth = numel(iSOUR.RegionLocationMinY0:iSOUR.RegionLocationMaxY1) * iSOUR.PhysicalDeltaY`Enter multiple values separated by semicolons, with no spaces.
- **left / right / top / bottom:** Define the boundaries of the actual ultrasound image within the larger DICOM frame. DICOM files often include large blank or text-filled areas. Determine coordinates manually for each preset using MATLAB:

- `image = dicomread('fullPathToDICOMFile');`
- `imagesc(image) colormap(gray)`

Use the *data tips* tool to identify the correct coordinates.

Enter values separated by semicolons, with no spaces.

- **summaryStatistic:** Specifies the summary statistic used for reporting ROI pixel values. Common choice: avg (mean). Other available statistics (always calculated and included in detailed export): sd, p05, p25, p50, p75, p95, mode.
- **normativeModelAvailable:** Set to yes when normative data and fitted models are available (see *Creating normative values*). Otherwise, set to no.
- **normativeModels:** When normative models are available, specify the filename of the MATLAB file containing them. Recommended: normModels.mat placed in the QUMIA installation folder.
- **bodyChartCoord:** Required for generating the schematic “Muscle Man” visualisation. This feature only functions when normative models are available, but the file must always be present. Recommended filename: bodyChartCoords.mat. See *Adjusting the Muscle Man* for editing instructions.
- **annotatedBodyIm:** Specifies the blank Muscle Man image used for annotation.

CREATING NORMATIVE VALUES

An example MATLAB script (*normModelsTemplate.m*) is provided to fit normative models to your dataset and automatically convert them into a format compatible with QUMIA. This chapter outlines the practical steps for generating these models using the recommended workflow, followed by an overview of the underlying statistical methodology.

1. Preparing the Normative Dataset

Create an Excel file containing the normative data in **long format**. The file must follow strict variable naming conventions:

- **ID:** Optional but recommended for traceability.
- **Sex:** Coded as M or F.
- **Age:** Numeric, in years.
- **Weight:** Numeric, in kilograms.
- **Height:** Numeric, in centimetres.
- **Muscle:** Muscle name, identical to the names listed in *Muscles.xlsx*.

When fitting **echogenicity** models, add:

- **EI:** Numeric, echogenicity in pixel grey value.

When fitting **muscle thickness** models, add:

- **TH:** Numeric, muscle thickness in centimetres.
- **Dominance:** Coded as 'Dominant' or 'Non-dominant'. Leave blank for muscles where handedness is irrelevant (all except upper-extremity muscles). Only complete this column when both sides have been measured.

Because the script requires long format, descriptive variables (age, sex, height, weight) must be repeated for each muscle entry.

2. Fitting the Models in MATLAB

- Open MATLAB and load *normModelsTemplate.m*.
- **For echogenicity models:**
 - Add the full path and filename of your normative dataset on **line 11**.
 - Run this section (Editor tab → *Run Section*, or press *Ctrl+Enter*).
 - The fitted models will be stored in the variable **Elmdls**.
- To visualise a model, specify the model number on **line 63** (e.g., *i = 1;*) and run that section.
- **For thickness models:**
 - Add the filename on **line 116**.
 - Run that section to generate the variable **THmdls**.
- To visualise thickness models, change the model number on **line 191** and run the section.
- Depending on which models you have fitted, select **Elmdls** and/or **THmdls** and save them as *normModels.mat* (recommended)
- Place this file in the QUMIA installation folder and update the *normModelAvailable* and *normModel* attributes in *QUMIA_config.txt* accordingly

Statistical Methodology

The normative modelling approach is based on data collected from the following population:

- Minimum of **80 healthy volunteers**
- Age range **0–80 years**
- **10 volunteers per 10-year age category**, ensuring adequate coverage
- Within each age category: **5 males and 5 females**
- Healthy volunteers defined as having **no neuromuscular disorder** or other pathology affecting muscle health
- Handedness, height and weight are **not controlled**

Pre-processing

Before model fitting, the following transformations are applied:

- Body mass index (BMI) is calculated.
- Age and BMI are **centred and scaled**.
- Age² and BMI² are computed, then also centred and scaled.

Model Fitting

A stepwise linear regression is performed using a **full factorial model** of all variables. Model selection is based on **Akaike's Information Criterion (AIC)**. This approach allows inclusion of covariates that may not be statistically significant individually but improve overall model performance. This is preferred for predictive modelling, where optimal fit is more important than strict hypothesis testing. The model is consecutively refitted, excluding outlier values. These are defined as having a Cook's distance more than 20 times the median Cook's distance.

ADJUSTING THE MUSCLE MAN

The muscles included in \Sitefiles_blank\Muscles.xlsx are already mapped onto the Muscle Man schematic. To add new muscles or modify existing muscle names, you must update their projections using *generateBodyChartCoords.m* in MATLAB.

Procedure

1. **Open MATLAB** and load *generateBodyChartCoords.m*.
2. Set the **QUMIA installation folder** as the current working directory.
3. On **line 7**, enter the muscle name for which you want to add or modify the projection.

Drawing or Reviewing a Projection

- **Run the first section** of the script.
 - If a projection already exists, it will be displayed in a figure window.
 - If the existing projection is correct, simply close the figure. MATLAB will display an error message; this can be safely ignored.
- **To add or modify a projection:**
 - Draw the new projection by clicking with the left mouse button to define each vertex of the polygon.
 - Complete the polygon by clicking again on the first point.
 - Ensure that the projection is drawn on the **right side** of the body.
 - The figure will close automatically once the polygon is completed.
- **To review the projection**, run the first section again.
 - If adjustments are needed, redraw the projection.
 - If the projection is correct, close the figure (again, MATLAB will throw an error that can be ignored).

Saving the Projection

Once the projection is correct:

- Run the **second section** of the script (starting at line 58).
- This saves the projection to the appropriate file.

Repeat this process for **each muscle** that requires a new or updated projection.